

# Customer Sentiment Analysis Case Study

End-to-End Data Analytics Project | SQL + Business Insights

## 1. Business Problem

An e-commerce company wants to understand customer behavior, satisfaction levels, and operational gaps across platforms to improve retention, service quality, and revenue growth.

## 2. Approach

Performed SQL-based analysis on customer sentiment dataset to evaluate purchasing trends, platform performance, complaint handling efficiency, and customer satisfaction metrics.

## 3. Key Insights Snapshot

| Area          | Insight                                |
|---------------|----------------------------------------|
| Age Group     | 18–35 contributes ~50%+ purchases      |
| Sentiment     | Positive ≈ Negative → Risk zone        |
| Top Platforms | Facebook Marketplace, Croma, Flipkart  |
| Complaints    | Paytm Mall & Myntra highest unresolved |
| Resolution    | Only 66.6% resolved                    |
| Trend         | Groceries & Automobiles rising demand  |

## 4. Deep Business Insights

• Young users (18–35) dominate → Digital-first strategy required. • Sentiment balance (~50-50) shows unstable customer satisfaction. • High-performing platforms drive majority of revenue. • Complaint inefficiency directly impacts trust and churn.

## 5. Strategic Recommendations

• Increase complaint resolution rate from 66% → 85%+ • Reduce response time (especially Myntra ~37 hrs) • Focus marketing on 18–35 segment • Improve service quality to shift sentiment positive • Expand inventory in trending categories

## 6. Expected Business Impact

• Higher customer retention • Improved brand trust • Increased repeat purchases • Revenue growth through better platform optimization

## 7. Tools & Skills Used

SQL, Data Analysis, Customer Segmentation, Sentiment Analysis, Business Storytelling